LESSON 6.2.4

Common AI Risks

A structured look at the risks AI introduces from biased outputs to autonomous actions and how governance, process, and people-centered controls can mitigate them.

RISK 1

Bias & Unfair Outcomes

Where Bias Comes From

- Historical inequalities in training data
- Subjective labeling practices
- Mismatched deployment context

How to Reduce It

- Define intended use clearly
- Establish fairness criteria upfront
- Evaluate performance by demographic group
- Document data provenance and known limitations

RISK 2

Accidental Data Leakage & Privacy

AI systems can inadvertently expose sensitive data even from anonymized sources.

🔍 Exposure Vectors

- PII embedded in training data
- Inputs logged for retraining
- Outputs that reconstruct individual details
- Re-identification of anonymized data

🛡️ Controls

- Data minimization
- Separate environments
- Disable data retention in approved tools
- Filtering, redaction, access controls
- Encrypt prompts, embeddings, and logs

📋 Governance

- Privacy impact assessments
- Required before training or handling production inputs

RISK 3

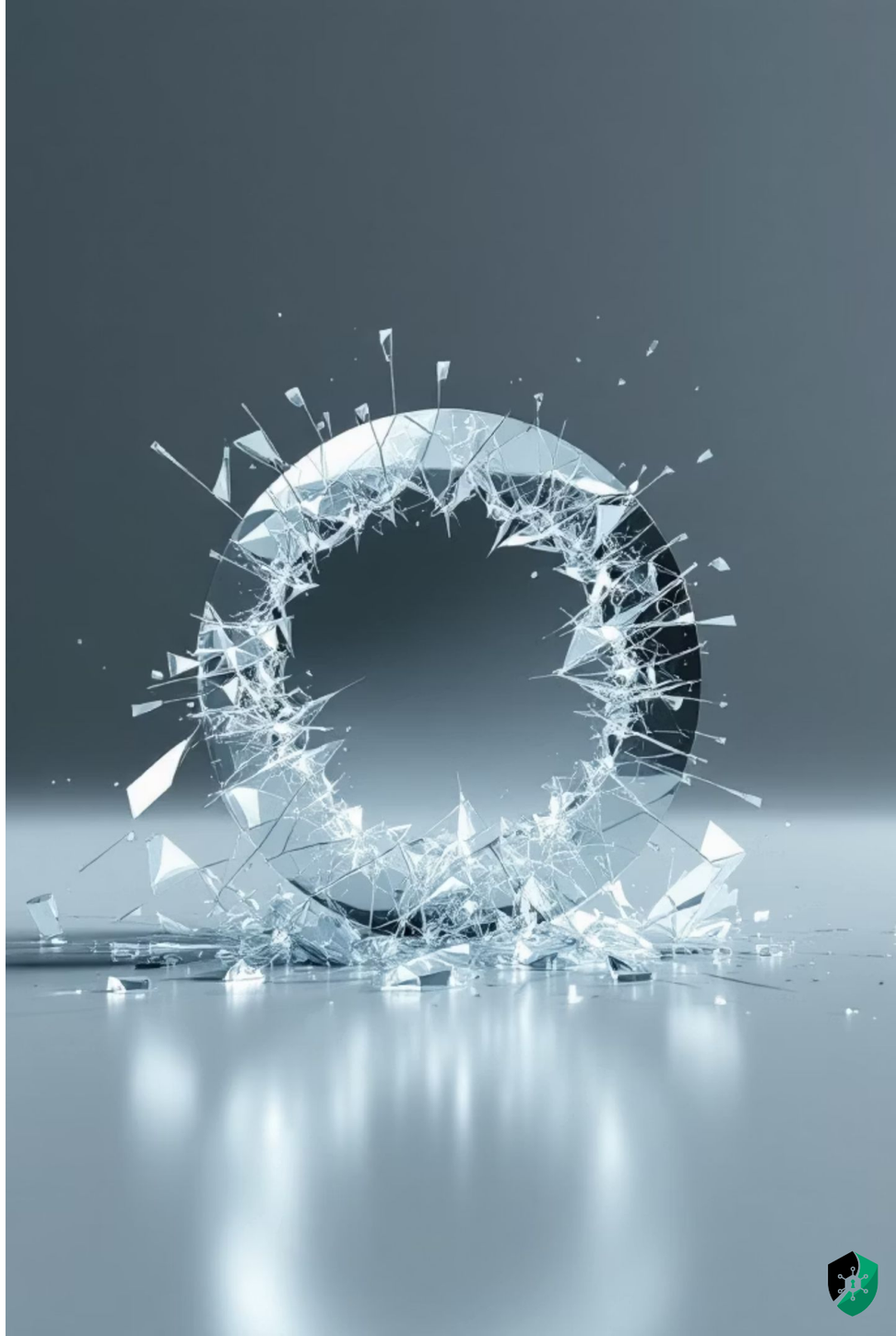
Reputational Harm & Stakeholder Trust

How Damage Occurs

- Offensive or inaccurate outputs
- Behavior inconsistent with brand values
- Lack of transparency about AI use

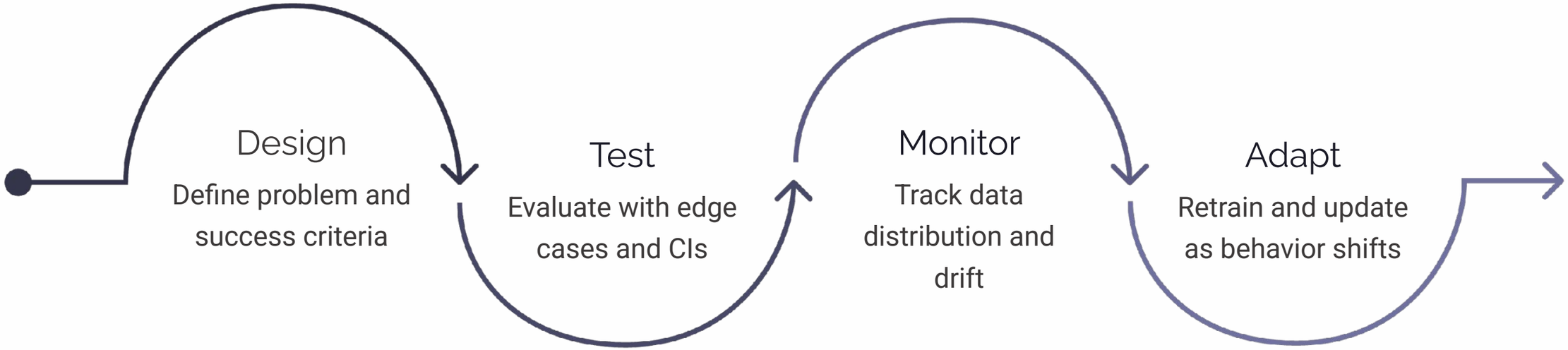
Governance Safeguards

- Human-in-the-loop escalation paths
- Product guidelines and response boundaries
- AI incident playbooks with rollback procedures
- Plain-language communication templates
- Transparent AI disclosures



RISK 4

Model Accuracy & Performance Drift



Performance degrades silently over time as data distributions change, adversaries adapt, or user behavior evolves. Proactive evaluation protocols are essential.

RISK 5

Intellectual Property & Content Provenance

Legal Risk

Training on copyrighted material without authorization; outputs may replicate protected content

Data Inventory

Catalog all training sources, verify licensing, maintain records for due diligence

Technical Controls

Watermarking, cryptographic signing of outputs, and IP checks embedded in review processes

RISK 6

Risks from Autonomy & Real-World Action

AI that can schedule tasks, make purchases, or control devices introduces new failure modes.

Limit Scope

Restrict actions to the minimum necessary; avoid open-ended permissions

Human Validation

Require approval for high-impact actions with context, previews, and justifications

Simulate First

Use dry-run and simulation modes before allowing live execution

Treat as Untrusted

Enforce authentication, authorization, and input validation on all API interactions

The AI Risk Landscape at a Glance

Bias
Unfair outcomes from skewed data

Privacy
PII leakage and re-identification

Reputation
Brand damage from harmful outputs

Drift
Accuracy loss over time

IP
Legal exposure from training data

Autonomy
Real-world errors from unchecked actions

Effective AI governance treats these risks as **interconnected**, spanning data, models, people, and processes.

Key Takeaways

AI is a socio-technical system

Risks stem from data and models, but also from people, processes, and environments

Bias requires proactive design

Define fairness criteria, evaluate by demographic, and document data provenance

Privacy controls must be layered

Minimization, separation, encryption, and impact assessments work together

Drift and accuracy need ongoing monitoring

Performance degrades — evaluation must be continuous, not one-time

Autonomous AI demands strict boundaries

Treat AI as an untrusted client; require human validation for high-impact actions